

Central limit theorem for frequencies of k -tuples in mutation systems

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DNA-Based Storage

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- In-vitro Vs. in-vivo.

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- The goal of this talk is to showcase a CLT on the frequency of k -tuples.
- We present a general definition of mutation systems and initial results on such systems.

Mutation Systems

- $\mathcal{A} = \{a_0, \dots, a_{d-1}\}$
- A word $w = w_0w_1 \dots w_{n-1}$ - finite concatenation of symbols over $\mathcal{A}$.
- $\mathcal{A}^k$ denotes the set of all k -tuples, $\mathcal{A}^* = \bigcup_{k \in \mathbb{N}} \mathcal{A}^k$.

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Definition

A **mutation law** is a pair $(\vartheta, \mathbf{P})$ where $\mathbf{P}$ is a set of d non-degenerate probability measures on $\mathcal{A}^*$, and $\vartheta : \mathcal{A} \rightarrow \mathcal{A}^*$ is a random function that corresponds to $\mathbf{P}$.

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Example

Let $\mathcal{A} = \{0, 1\}$ and define

$$\vartheta(0) = \begin{cases} 00 & \text{w.p. } \frac{2}{3} \\ 01 & \text{w.p. } \frac{1}{3} \end{cases}, \quad \vartheta(1) = \begin{cases} 1 & \text{w.p. } \frac{1}{4} \\ 10 & \text{w.p. } \frac{1}{4} \\ 110 & \text{w.p. } \frac{1}{2} \end{cases}$$

Mutation Systems

Definition

Let $w = w_0 \dots w_{m-1}$ be a word. A **mutation step/round** is performed by first selecting a position $i \in [m]$ uniformly at random, and then replacing the symbol w_i at the selected position with $\vartheta(w_i)$. With a slight abuse of notation, we denote the obtained random word by $\vartheta(w)$.

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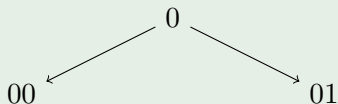
Definition

A **mutation system** is defined by $S = (\mathcal{A}, w, (\vartheta, \mathbf{P}))$ where $w \in \mathcal{A}^*$ is a non-empty starting word. A mutation system evolves according to the discrete random model defined by mutation rounds $\vartheta^n(w)$.

Mutation Systems

Example

Let $\mathcal{A} = \{0, 1\}$ and assume our mutation is a single noisy duplication: $\vartheta(0) = 00$ w.p. $3/4$ and 01 w.p. $1/4$ (and similarly for $\vartheta(1)$). Assume also that $w = 0$. Then

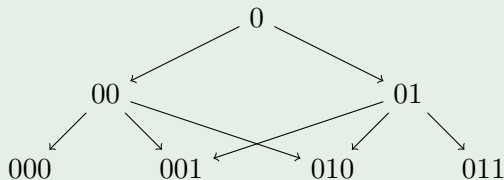


$$0 \xrightarrow[\text{w.p. } 3/4]{\vartheta(0)=00} 00, \quad 0 \xrightarrow[\text{w.p. } 1/4]{\vartheta(0)=01} 01$$

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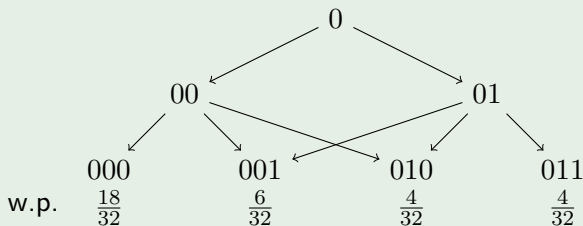


$$00 \xrightarrow[\text{w.p. } \frac{1}{2} \cdot \frac{3}{4}]{\vartheta(0)0} 000, \quad 00 \xrightarrow[\text{w.p. } \frac{1}{2} \cdot \frac{3}{4}]{0\vartheta(0)} 000, \quad 00 \xrightarrow[\text{w.p. } \frac{1}{2} \cdot \frac{1}{4}]{\vartheta(0)0} 010$$

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Related Papers

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- Most related paper - "On the Long-Term Behavior of k -tuples Frequencies in Mutation Systems"[E '24]. In which the limit frequency (in probability) of k -tuples was found.
- In this work we further extend the limit results into a CLT, using similar tools (martingale convergence theorem).

Studying the Long-Term Behavior

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Definition

- ▶ For two words $u, v \in \mathcal{A}^*$, we denote by $\text{ct}_v(u)$ the number of times that u appears in v , where coordinates are taken modulo $|v|$.
- ▶ Similarly, we denote by $\text{fr}_v(u)$ the empirical frequency of u in v , i.e., $\text{fr}_v(u) = \text{ct}_v(u)/|v|$.
- ▶ For $k \in \mathbb{Z}$, we denote by $\text{ct}_v^{(k)}$ and $\text{fr}_v^{(k)}$ the count and frequency vectors of all possible k -tuples ordered in the lexicographic order.

The Substitution Matrix

- Our next objective is to present the mutation law in a matrix form that incorporates key information about $\mathbf{fr}^{(k)}$, enabling us to apply algebraic tools.
- Since the general definition (for general k) is less intuitive, we present the matrix for $k = 1$.

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Definition

Let $(\vartheta, \mathbf{P})$ be a mutation law over an alphabet $\mathcal{A}$ of size d . The **substitution matrix** $\mathbf{M}$ is a real, $d \times d$ matrix, with non-negative entries, defined by

$$\mathbf{M}_{i,j} := \mathbb{E}[\text{ct}_{\vartheta(a_j)}(a_i)] = \sum_{\eta \in \mathcal{A}^*} \Pr(\vartheta(a_j) = \eta) \text{ct}_{\eta}(a_i).$$

-For general k , $\mathbf{M}^{(k)}$ is a $d^k \times d^k$ matrix constructed similarly.

Average τ -Mutation Systems

Example

Let $\mathcal{A} = \{0, 1\}$ be the binary alphabet. Consider the duplication law

$$\vartheta(0) = \begin{cases} 00 & \text{w.p. } 2/3 \\ 01 & \text{w.p. } 1/3 \end{cases}, \quad \vartheta(1) = \begin{cases} 11 & \text{w.p. } 3/4 \\ 00 & \text{w.p. } 1/4. \end{cases}$$

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Calculating the $(0, 0)$ entry in $\mathbf{M}$ requires finding the expected number of zeros in $\vartheta(0)$. We obtain that $\mathbb{E}[\text{ct}_{\vartheta(0)}(0)] = 2 \cdot 2/3 + 1 \cdot 1/3 = 5/3$.

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$$\mathbf{M} = \begin{bmatrix} 5/3 & 1/2 \\ 1/3 & 3/2 \end{bmatrix}.$$

Average τ -Mutation Systems

Definition

A mutation system $S = (\mathcal{A}, w, (\vartheta, \mathbf{P}))$ is called an **Average τ -Mutation System** if $\exists \tau \in \mathbb{N}$ such that $\mathbb{E}[|\vartheta(a)|] = \tau$ for all $a \in \mathcal{A}$. This is equivalent to requiring every column of $\mathbf{M}$ to sum to τ .

Theorem (E 24')

Let $S = (\mathcal{A}, w, (\vartheta, \mathbf{P}))$ be an Average τ -Mutation System. Assume its substitution matrix $\mathbf{M}^{(k)}$ is irreducible. From Perron-Frobenius theorem we know there is a single maximal eigenvalue of $\mathbf{M}^{(k)}$ with a single eigenvector $\mathbf{r}$ normalized to be a probability vector (sum of entries is one), assuming a regular eigendecomposition exists. Then

$$\mathbf{fr}_n^{(k)} \xrightarrow{\text{Pr}} \mathbf{r}.$$

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We get $\tau = 2$ and a unique maximal eigenvalue $\lambda_0 = 2$, $\mathbf{r} = \begin{bmatrix} 3/5 & 2/5 \end{bmatrix}^T$. Assume the starting word is w . We get

$$\mathbf{fr}_n \rightarrow \begin{bmatrix} 3/5 \\ 2/5 \end{bmatrix}.$$

Central Limit Theory Over Average τ -Mutation

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Theorem (Central Limit Theorem for Average τ -Mutation)

Let ϑ be a mutation on a starting word w in an average τ -mutation system. If $\mathbf{M}^{(k)}$ is irreducible with a unique maximal eigenvalue and normalized eigenvector $\mathbf{r}$, then under mild additional assumptions:

$$\sqrt{n} \left(\frac{\mathbf{ct}_n^{(k)}}{n} - (\tau - 1) \mathbf{r} \right) \xrightarrow{d} N(\mathbf{0}, \Sigma)$$

for some computable covariance matrix Σ .

Outline of the Proof

- We construct a matrix U row-by-row where each row is an eigenvector of $\mathbf{M}^{(k)}$ (separated to their real and imaginary parts), $\mathbf{u}_i^r$ and $\mathbf{u}_i^c$.

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- We project the asymptotically centered vector $\mathbf{c}\mathbf{t}_n^{(k)} - n(\tau - 1)\mathbf{r}$ over the rows of U , and obtain the vector $\mathbf{X}_n$ of projections. We show the asymptotic jointly normal distribution of $\mathbf{X}_n$.

$$\frac{1}{\sqrt{n}}\mathbf{X}_n = U\sqrt{n}\left(\frac{\mathbf{c}\mathbf{t}_n^{(k)}}{n} - (\tau - 1)\mathbf{r}\right) \xrightarrow{d} N(\mathbf{0}, \Sigma').$$

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- This is done by showing the central limit theorem for every linear sum of entries with weights α_i

$$\frac{1}{\sqrt{n}} \sum_{i=0}^{d^k-1} \alpha_i X_n^{(i)} \xrightarrow{d} N(\mathbf{0}, \sigma^2).$$

Outline of the Proof

- We define a martingale differences process $M_n^{(i)}$ using the entries of $\mathbf{X}_n$, the length of the mutation in the n 'th step Y_n and the i 'th eigenvalue of the substitution matrix λ_i ,

$$M_n^{(i)} := X_n^{(i)} - X_{n-1}^{(i)} - \frac{1}{Y_{n-1}} (\lambda_i - k) X_{n-1}^{(i)}.$$

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- We will approximate the sum on $X_n^{(i)}$ as a sum of $M_n^{(i)}$ with coefficients $\beta_{j,n}^{(i)}$

$$\frac{1}{\sqrt{n}} \sum_{i=0}^{d^k-1} \alpha_i X_n^{(i)} \stackrel{L^1}{\sim} \sum_{j=1}^n \frac{1}{\sqrt{n}} \sum_{i=0}^{d^k-1} \beta_{j,n}^{(i)} M_j^{(i)}.$$

Outline of the Proof

- Now we show CLT over the sum of martingale differences processes by using CLT for Martingale and proving its conditions hold, showing

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- Proving the CLT for the sum of martingales proves the CLT for the counting vector and thus proves the theorem. Lastly we use the value of σ^2 to calculate the covariance matrix Σ and we finish the proof

$$\sqrt{n} \left(\frac{\text{ct}_n^{(k)}}{n} - (\tau - 1) \mathbf{r} \right) \xrightarrow{d} N(\mathbf{0}, \Sigma)$$

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Calculating the CLT we find the complete asymptotical behavior of the symbols in the mutation:

$$\sqrt{n} \left(\frac{\mathbf{ct}_n^{(1)}}{n} - \begin{bmatrix} 3/5 \\ 2/5 \end{bmatrix} \right) \xrightarrow{d} N(\mathbf{0}, \Sigma)$$

with Σ the covariance matrix:

$$\Sigma = \begin{bmatrix} 2/3 & -2/3 \\ -2/3 & 2/3 \end{bmatrix}.$$

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- Knowing the asymptotic distribution can allow us to confirm a suggested mutation law of a system.
- These results provide a deeper insight into the long-term process of DNA mutations which is crucial to ensuring the reliability and efficiency of DNA data storage systems.

Future Problems

- ① Proving CLT for mutations over blocks of symbols instead of single symbols
- ② Finding effective error-correcting coding schemes

Thank you!